

ARITRI HALDER

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Mechanical Engineer | Thermal Systems | Prototype Builds | Test & Validation | NPD/NPI | Data Analysis

SUMMARY

Ph.D. candidate in Mechanical Engineering with 3+ years of hands-on R&D; experience with DFMA, V&V, NPD. Designed and tested a 20 kW high-temperature industrial burner, integrated sensors and controls, validated system performance, and used experimental data to guide engineering decisions and achieved 60% reduction in NOx emissions compared to conventional burners.

EDUCATION

University of Minnesota - Twin Cities | Minneapolis, MN | Expected June 2027

Ph.D., Mechanical Engineering

Jadavpur University | Kolkata, India | July 2023

B.E., Mechanical Engineering

EXPERIENCE

University of Minnesota - T.E. Murphy Engine Lab | Graduate Research Assistant

Minneapolis, MN | Oct 2024 - Present

- Designed and validated a 20 kW high-temperature furnace and modular test hardware, delivering repeatable process data and engineering recommendations with CenterPoint Energy and U of M Duluth; achieved 60% reduction in NOx.
- Developed SolidWorks models, GD&T; drawings, tolerance stack-ups, and BOMs; applied DFM/DFA, fit/clearance analysis, and material selection for prototypes operating near 1000 deg C.
- Engineered and commissioned an ammonia/methane/air process system integrating mass flow controllers, sensors, NI cDAQ, and LabVIEW for automated control and synchronized data acquisition.
- Planned a four-variable DOE and applied statistical analysis, FTIR, PIV, chemiluminescence, and CT mechanisms to define operating windows, validate performance, and characterize NOx/ammonia trends.
- Built a single-camera rotational CT system and automated Python/MATLAB processing of 200-300 images per condition; applied RCA and FMEA to improve reliability and safety.

IIT Kanpur - Experimental Thermofluids Laboratory | Research Intern

Kanpur, India | May 2022 - Jul 2022

- Ran parameterized study of mesh convergence to verify pressure results were independent of mesh size in ANSYS Fluent.
- Modified internal insert geometry and CFD simulation results improved minichannel thermal-fluid performance by up to 28%.

Jadavpur University - Microfluidics Lab | Research Assistant

Kolkata, India | Aug 2022 - Jun 2023

- Developed and executed parallel C/C++ simulations on an HPC system to characterize unsteady fluid-structure interaction at $Re = 250$.
- Quantified wake dynamics, vorticity, and flow separation across Strouhal number and pitching phase in Tecplot.

SELECTED PUBLICATION & PRESENTATIONS

- A. Halder, C. Goertemiller, A. Hoxie, and W. F. Northrop, "Experimental investigation of a 20 kW atmospheric pressure pure ammonia jet-swirl burner," Proceedings of the Combustion Institute, 2026.
- A. Halder, R. Pandey, and U. Madanan, "Computer simulations on heat transfer enhancement in a minichannel heat sink using modified butterfly inserts," ISHMT-ASTFE Heat and Mass Transfer Conference, pp. 149-154, 2024.
- A. Halder et al., "Stability and Emissions Profile of a 120 kW Fuel-Flexible Jet-Swirl Burner Operating on Ammonia." In progress; Transformative Energy.
- Five oral/poster presentations at international and U.S. technical meetings, including ISOC Kyoto (2026), 14th U.S. National Combustion Meeting (2025), and 4th Symposium on Ammonia Energy (2025).

SKILLS

Quality & Continuous Improvement: FMEA, RCA, DOE, corrective-action problem solving, continuous improvement, process characterization, statistical analysis, process controls, SOPs, calibration and measurement-system validation.

Mechanical Design & Manufacturing: SolidWorks 3D modeling, engineering drawings, GD&T;, DFM/DFA, tolerance stack-up, fit/clearance analysis, BOMs, materials selection, prototyping, AutoCAD, Autodesk Fusion 360.

Instrumentation, Test & Characterization: NI cDAQ, LabVIEW, mass flow controllers, pressure/temperature sensors, sensor integration, calibration, synchronized data acquisition, test automation, PIV, FTIR spectroscopy, optical diagnostics, chemiluminescence imaging, CTC.

Data Analysis & Technical Reporting: Python, MATLAB, C/C++, Microsoft Excel, PowerPoint, Word, Tecplot, image processing, workflow automation, data visualization, statistical analysis, technical reporting.

Modeling & Simulation: ANSYS Fluent, FVM, FEM, CFD, heat transfer, thermal-fluid modeling, numerical methods, mesh convergence, HPC, Cantera.

AWARDS

- Chester E. Dekko Fdn Endowed Graduate Fellowship, University of Minnesota, 2024-2025.
- Best Abstract - Thermal Applications, 4th Symposium on Ammonia Energy, 2025.